Fundamentals of Artificial Intelligence and Knowledge Representation – Module 4

Prof. Federico Chesani – 28th of June, 2021

Available time: 25 min. + 10 min. for submitting the essay

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FamilyNameFirstName.date.pdf/.docx
For example, if you are John Smith, the file will be named:
SmithJohn.20210628.pdf

Exam A

1) The candidate is invited to write a prolog predicate **`count/2`** that, given as input a term representing a predicate, it will return the number of solutions for the goal of proving that input predicate. For simplicity, it will be assumed that the proof procedure of the predicate passed as parameter will always terminate in a finite time.
For example:

```
p(a).
p(b).
p(c).
```

and the query:

```
count(p(X), Result).
```

The expected answer is:

```
Yes, Result = 3.
```

This is because there are three different ways for proving the goal **`p(X).`**

2) The candidate is invited to introduce the RETE algorithm, and to show with a short example in natural language the differences between inter-elements and intra-elements patterns/features.